Python for AI & Machine Learning

Class 22 – Introduction to Pandas

Course: Python for AI & Machine Learning
Module: Data Analysis with Python
Duration: 2 Hours (Theory + Practical)

Learning Objectives

By the end of this session, students will be able to:

- Understand what Pandas is and why it is used.
- Install and import the Pandas library.
- Understand the concepts of Series and DataFrames.
- Create Series and DataFrames from different data sources.
- Read CSV and Excel files using Pandas.
- Explore and inspect datasets.
- Select and filter data.
- Understand the role of Pandas in AI, Machine Learning, and Data Science.

Revision

In the previous class, we learned about **Advanced NumPy Operations**, including:

- Array Reshaping
- Flattening Arrays
- Resizing Arrays
- Broadcasting
- Boolean Indexing
- Fancy Indexing
- Sorting and Searching Arrays
- Stacking and Splitting Arrays
- Vectorization

Today, we will begin learning **Pandas**, one of the most powerful libraries for data analysis and preprocessing.

Introduction to Pandas

Pandas is an open-source Python library used for:

- Data Analysis
- Data Manipulation
- Data Cleaning
- Data Visualization Preparation
- Machine Learning Data Preprocessing

The name **Pandas** comes from **Panel Data**, which refers to multidimensional structured datasets.

Pandas is built on top of the **NumPy** library, making it fast and efficient for handling structured data.

Why Do We Need Pandas?

Imagine you have:

- Student records
- Employee information
- Hospital data
- Banking transactions
- Sales reports
- Machine Learning datasets

Managing thousands or millions of records using Python lists or dictionaries is difficult.

Pandas makes it easy to:

- Read datasets
- Organize data
- Search records
- Filter information
- Analyze trends
- Prepare data for Machine Learning models

Features of Pandas

- High-speed data processing
- Easy data manipulation
- Built-in data cleaning tools
- Supports CSV, Excel, JSON, SQL databases
- Missing value handling
- Data filtering
- Data grouping
- Statistical analysis

- Seamless integration with NumPy and Matplotlib

Installing Pandas

Install Pandas using pip:

```bash id="p7f9x1" pip install pandas

````
Check the installation:

```bash id="p4g8y2"
pip show pandas
````

Importing Pandas

```python id="p3d6a7" import pandas as pd

````
`pd` is the standard alias used for the Pandas library.

What is a Series?

A **Series** is a one-dimensional labeled array.

It is similar to a Python list but includes an index.

Creating a Series

```python id="p8k2m4"
import pandas as pd

marks = pd.Series([85, 90, 78, 95, 88])

print(marks)
````

Output

```text id="p5q3t8" 0 85 1 90 2 78 3 95 4 88 dtype: int64
```

````
---

# Creating a Series with Custom Index

```python id="p9m7w6"
import pandas as pd

marks = pd.Series(
    [85, 90, 78],
    index=["Rahul", "Sneha", "Aman"]
)

print(marks)
````

Output

```text id="p2l8z9" Rahul 85 Sneha 90 Aman 78 dtype: int64

````

What is a DataFrame?

A **DataFrame** is a two-dimensional table consisting of rows and columns.

It is similar to:

- Excel Spreadsheet
- SQL Table
- Database Table

The DataFrame is the most commonly used data structure in Pandas.

Creating a DataFrame

```python id="p1b4c5"
import pandas as pd

student = {
    "Name": ["Rahul", "Sneha", "Aman"],
    "Age": [22, 21, 23],
    "Marks": [90, 95, 88]
}

df = pd.DataFrame(student)

print(df)
````
````

Output

```text id="p6d7e8" Name Age Marks 0 Rahul 22 90 1 Sneha 21 95 2 Aman 23 88

````

Understanding Rows and Columns

In the above DataFrame:

Rows:

- Rahul
- Sneha
- Aman

Columns:

- Name
- Age
- Marks

Reading a CSV File

CSV stands for **Comma-Separated Values**.

```python id="p0r4f6"
import pandas as pd

df = pd.read_csv("students.csv")

print(df)
````

CSV files are widely used in AI and Machine Learning datasets.

Reading an Excel File

```python id="p5k2a3" import pandas as pd

df = pd.read_excel("students.xlsx")

print(df)
```

````
---

# Viewing the First Records

```python id="p8d6h9"
print(df.head())
````

Displays the first five rows.

Viewing the Last Records

```python id="p4g9t2" print(df.tail())

````
Displays the last five rows.

Checking Dataset Information

```python id="p3w8m5"
print(df.info())
````

Shows:

- Number of rows
- Number of columns
- Data types
- Missing values

Statistical Summary

```python id="p9n4y6" print(df.describe())

````
Provides:

- Count
- Mean
- Standard Deviation
- Minimum
- Maximum
- Quartiles

````

````
# Displaying Column Names

```python id="p6z1r8"
print(df.columns)
````

Checking Dataset Shape

```python id="p1v7t9" print(df.shape)

````
Output

```text id="p2w3x4"
(100, 5)
````

This means:

- 100 Rows
- 5 Columns

Selecting a Single Column

```python id="p7k5j1" print(df["Marks"])

````

Selecting Multiple Columns

```python id="p3m9c8"
print(df[["Name", "Marks"]])
````

Selecting Rows using `loc`

```python id="p2a8d7" print(df.loc[0])

````
Returns the first row.

Selecting Rows using `iloc`
````

````
```python id="p4q1v6"
print(df.iloc[2])
````

Returns the third row using index position.

Filtering Data

Display students who scored more than 90.

```python id="p5t8l2" print(df[df["Marks"] > 90])

````

Sorting Data

Sort students based on marks.

```python id="p9y4w7"
print(df.sort_values("Marks"))
````

Descending order:

```python id="p8x3e6" print(df.sort_values("Marks", ascending=False))

````

AI Example – Student Dataset

```python id="p6n2f4"
import pandas as pd

students = {
    "Name": ["Rahul", "Sneha", "Aman"],
    "Marks": [92, 84, 97]
}

df = pd.DataFrame(students)

print(df[df["Marks"] >= 90])
````

````

AI Example – Employee Records

```python id="p7r3u5" import pandas as pd

employees = { "Employee": ["John", "Sara", "David"], "Salary": [45000, 52000, 61000] }

df = pd.DataFrame(employees)

print(df.describe())

```

AI Example – Sales Analysis

```python id="p4j8m9"
import pandas as pd

sales = {
    "Month": ["Jan", "Feb", "Mar"],
    "Revenue": [50000, 62000, 58000]
}

df = pd.DataFrame(sales)

print(df.sort_values("Revenue", ascending=False))
```

Applications of Pandas

Pandas is widely used in:

- Data Science
- Machine Learning
- Artificial Intelligence
- Business Intelligence
- Financial Analysis
- Healthcare Analytics
- Banking Systems
- Sales Reporting
- Research Projects

Best Practices

- Use meaningful column names.
- Keep data clean and organized.

- Verify data types before analysis.
- Handle missing values properly.
- Use DataFrames instead of multiple lists for structured data.
- Always inspect datasets before building Machine Learning models.

Practical Exercise 1

Create a Series containing:

- Student Names
- Student Marks

Display the Series.

Practical Exercise 2

Create a DataFrame containing:

- Name
- Age
- Course
- Marks

Display the DataFrame.

Practical Exercise 3

Read a CSV file.

Display:

- First five rows
- Last five rows
- Shape
- Column names

Practical Exercise 4

Filter students who scored above 80.

Sort the result in descending order.

Practical Exercise 5

Create an Employee DataFrame.

Display:

- Average Salary
- Highest Salary
- Lowest Salary

Assignment

AI Student Performance Dashboard

Create a Pandas application that:

- Reads a CSV file containing student records.
- Displays the first five rows.
- Displays the last five rows.
- Shows dataset information.
- Displays statistical summary.
- Filters students scoring above 75.
- Sorts students based on marks.
- Displays the total number of students.

Bonus Challenge:

Add a new column called **Grade** based on marks:

- A → 90 and above
- B → 75–89
- C → Below 75

Interview Questions

1. What is Pandas?
2. Why is Pandas important in Data Science?
3. What is a Series?
4. What is a DataFrame?
5. What is the difference between a Series and a DataFrame?
6. How do you read a CSV file using Pandas?
7. What is the difference between `loc` and `iloc`?
8. How do you filter data in Pandas?
9. What is the purpose of `head()` and `tail()`?
10. How is Pandas used in AI and Machine Learning?

Summary

In today's session, we learned:

- Introduction to Pandas
- Installing and Importing Pandas
- Series
- DataFrames
- Reading CSV and Excel Files
- Exploring Datasets
- Selecting Rows and Columns
- Filtering Data
- Sorting Data
- AI & Machine Learning Applications
- Practical Exercises
- Assignment

Pandas is one of the most essential libraries in Python for handling structured data. It simplifies data cleaning, analysis, and preparation, making it a core tool in Data Science, Artificial Intelligence, and Machine Learning.

Next Class Preview

Class 23 – Data Cleaning and Data Preprocessing with Pandas

Topics to be covered:

- Understanding Missing Data
- Handling Missing Values (`isnull()`, `dropna()`, `fillna()`)
- Removing Duplicate Records
- Renaming Columns
- Changing Data Types
- String Operations
- Applying Functions with `apply()`
- Data Transformation
- Preparing Datasets for Machine Learning